

Haley Colgate Kottler, PhD(c)

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Education and Certifications

University of Wisconsin - Madison | PhD in Mathematics

Spring 2026

University of Wisconsin - Madison | MA in Mathematics

2022

Colorado College | BA in Mathematics

2020

Skills

Languages and platforms: R, Python, Julia, SQL, LaTeX, Linux, Excel

Quantitative: Causal inference, Machine learning, Reinforcement learning, Statistical modeling

Soft skills: Presenting, Writing, Adapting, Mentoring

Certifications: [Erdős Institute Data Science Bootcamp](#) Spring 2025

Professional Experience

Lead Data Analyst

2024-present

BerbeeWalsh Department of Emergency Medicine | Madison, WI

- Designed data processing pipelines in R for novel multilevel data leading to two research papers
- Progressed study design and analysis plans with preliminary data for grant applications to fund the continuation of a multi-year collaborative research program
- Mentored three junior colleagues to guide them in determining appropriate statistical techniques for independent research projects and presenting results to wide audiences

Mathematics Instructor

2023

University of Wisconsin-Madison | Madison, WI

- Devised assessments, lecture notes, and course policies for an accelerated summer semester class to train primarily engineering students on strategies for modeling real-world applications using differential equations
- Overhauled lectures and in class activities for 60 students in a developmental math class using principals of backwards design to ensure they are prepared for the math fundamentals necessary for entry level courses
- Developed scaffolded exercises and corresponding assessments to assess student understanding

Mathematics Teaching Assistant

2020-2022

University of Wisconsin-Madison | Madison, WI

- Established a shared resource bank for course materials and coordinated with colleagues to streamline workflows
- Provided constructive feedback to students to promote recognition of areas for growth
- Advised junior colleagues on strategies for addressing the root cause of why students are struggling

Leadership Experience

Outreach Coordinator | Mega Math Meet | Madison, WI

2023-present

- Directed grading and logistics for an annual mathematics competition for 180 fifth and sixth grade students and developing problems based on advanced concepts not included in the usual K12 math curriculum

Program Committee Member | University of Wisconsin-Madison

2023-present

- Consolidated and presented the wishes of the mathematics graduate student body to department leadership to guide improvement and future directions of the program

Co-founder | UGMAAW | University of Wisconsin-Madison

2023-2025

- Founded a community among undergraduate students studying mathematics who may not feel included by creating space where they can find support from their peers and advance themselves professionally

Selected Publications

[Method of moments for Gaussian mixtures](#)

November 2025

[A proxy-based approach for overcoming unmeasured confounding in EHR studies](#)

June 2025

[Characterizing Lucid Episodes in Advanced Dementia](#)

Aug 2025

[Quantifying aberrant approach-avoidance conflict in psychopathology](#)

April 2023

[Batch Codes from Affine Cartesian Codes and Quotient Spaces](#)

Dec 2021

[Colorado in context: Congressional redistricting and competing fairness criteria in Colorado](#)

May 2021